

## POWER MARKETS

# Casebook Q&A Overview: Applied Decision Cases

The casebook turns the lectures into bounded decisions: quote a product, size storage, assess a site, audit a partner, and choose an operating model under incomplete information.

# What does each case produce?

|           |                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What makes the casebook different from another summary chapter?                                                                                                                                               |
| ANSWER    | Each case starts from a commercial situation and ends in a concrete artifact: a quote stack, duration hurdle, site scorecard, revenue-quality ladder, partner scorecard, product test, or capability roadmap. |
| DEEPEN IN | Cases C01-C07; Vol II Ch 7, Ch 16; Vol III Ch 12.                                                                                                                                                             |

1

## Situation

- A site, customer, battery, partner, portfolio, or operating-model question has incomplete data and several plausible paths.

2

## Work product

- The answer has to be written as something a team could review, challenge, and improve.

3

## Stress test

- One assumption changes after the recommendation.
- The case is useful when the change exposes the hinge variable.

4

## Revision

- The artifact improves by changing evidence, assumptions, constraints, and decision boundaries.

# What decision does each case stress?

QUESTION

ANSWER

DEEPEN IN

How do the seven cases cover the applied layer of the course?

The cases cover recurring renewable-IPP decisions where physical limits, market structure, contract promises, and governance meet.

Cases C01-C07.

## Case map

| Case | Decision object                 | Primary artifact       | Main constraint                                          |
|------|---------------------------------|------------------------|----------------------------------------------------------|
| C01  | Shaped solar+BESS PPA           | Quote stack            | Customer promise versus residual exposure                |
| C02  | 1h / 2h / 4h duration           | Duration hurdle        | Marginal MWh value versus constraints                    |
| C03  | Grid-constrained site           | Site scorecard         | Grid rights can dominate resource quality                |
| C04  | BESS stack under saturation     | Revenue-quality ladder | One asset spends the same MW, MWh, SoC, and cycle budget |
| C05  | BRP/BSP performance             | Partner scorecard      | Execution needs a feasible benchmark                     |
| C06  | Data-centre clean-power product | Product boundary test  | Hourly residual risk and evidence                        |
| C07  | Year-one operating model        | Capability roadmap     | Build only what the portfolio can govern                 |

# C01: how is a shaped solar+BESS PPA quoted?

|           |                                                                                                                                                                                      |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Where is the offer boundary when a customer asks for smoother clean supply than raw solar?                                                                                           |
| ANSWER    | The seller has to price the transformation from production shape to customer promise and avoid residual risks that remain unbounded or uncontrolled.                                 |
| ANCHOR    | Scenario quote is 65 EUR/MWh versus a buyer ceiling of 68 EUR/MWh, but the stress-hour residual is 0.60m EUR/year before caps, collars, and annual clean-energy shortfall treatment. |
| RELATION  | Quote stack = raw energy + shaping/firming + imbalance/grid + evidence/credit + residual-risk margin.                                                                                |
| DEEPEN IN | Case C01; Vol II Ch 4, Ch 5, Ch 6, Ch 7; Vol III Ch 5.                                                                                                                               |

1

## Define the promise

- Shape, settlement basis, certificates, data obligations, tolerance bands, and fallback rules.

2

## Estimate residual exposure

- Solar, BESS, portfolio support, and market procurement reduce hard-hour risk while leaving a residual tail.

3

## Price the stack

- Energy, shape, imbalance, grid, evidence, credit, optionality, and margin.

4

## Set the boundary

- The offer breaks when residual risk, control rights, or fallback obligations become too open-ended for the price.

# C02: when is extra storage duration justified?

|           |                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How are 1h, 2h, and 4h BESS options compared?                                                                                               |
| ANSWER    | Extra duration is a marginal investment with its own scarce hours, capex, degradation, warranty, insurance, tariff, and product-depth test. |
| ANCHOR    | In the case, 2h adds about 0.19m EUR/year after hurdle while 4h is about -0.26m EUR/year before stronger evidence for longer scarcity.      |
| RELATION  | Required energy = MW obligation × duration, but the investment case depends on marginal net revenue and constraints.                        |
| DEEPEN IN | Case C02; Vol I Ch 13; Vol II Ch 9, Ch 12; Vol III Ch 6.                                                                                    |

1

## Find the marginal scarcity

- Identify which hours require additional MWh after shorter duration is exhausted or reserved.

2

## Price the incremental asset

- Include capex, augmentation, degradation, grid impact, warranty and insurance restrictions, and opportunity cost.

3

## Test market depth

- Competition and rule changes can make a real product too shallow for the extra capacity.

4

## Decide

- Extra duration is attractive only when the marginal MWh clears its own hurdle under realistic operating limits.

## C03: which grid-constrained site advances?

|           |                                                                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How can a lower-resource site outrank a higher-resource site?                                                                                  |
| ANSWER    | The investable site is the one whose resource, connection, curtailment, tariff, and optionality package creates the better risk-adjusted path. |
| RELATION  | Site value changes with delivered MWh, curtailment, connection timing, tariff structure, and flexibility rights.                               |
| DEEPEN IN | Case C03; Vol I Ch 4; Vol II Ch 2; Vol III Ch 3.                                                                                               |

1

### Separate resource from grid right

- Irradiation or wind quality is only one input.

2

### Score the wire side

- Export/import capacity, queue risk, curtailment, connection cost, tariffs, losses, and local flexibility.

3

### Test co-location

- Storage can improve or worsen the constraint depending on grid rights and operating mode.

4

### Rank the sites

- A weaker resource with a better connection can beat a stronger resource behind costly or uncertain access.

## C04: how robust is a BESS stack under saturation?

|           |                                                                                                                                      |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What happens when many batteries chase similar revenue products?                                                                     |
| ANSWER    | The stack has to be tested for mutually exclusive use of MW, MWh, SoC, cycle budget, grid rights, qualification, and control rights. |
| ANCHOR    | The same MW, MWh, state-of-charge headroom, and cycle budget cannot be spent twice.                                                  |
| DEEPEN IN | Case C04; Vol II Ch 8, Ch 10, Ch 12; Vol III Ch 1.                                                                                   |

1

### Candidate revenues

- Arbitrage, reserve capacity, balancing energy, co-location, grid service, customer product, and optionality.

2

### Scarce-resource map

- Each line claims physical capability or operating rights from the same asset.

3

### Revenue quality

- Separate underwritten floor, repeatable market value, optimizer forecast, merchant upside, and option value.

4

### Decision hinge

- A revenue line can be real and still too small, too crowded, or too operationally expensive to carry the investment case.

## C05: how is BRP/BSP performance audited?

|           |                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How can an asset owner judge outsourced execution without hindsight bias?                                                                    |
| ANSWER    | The audit needs raw data and a feasible counterfactual built from the information, rights, liquidity, and constraints available at the time. |
| ANCHOR    | The headline gap is 0.50m EUR/year, but only about 0.11m EUR/year remains unresolved after feasible benchmarking and data gaps.              |
| DEEPEN IN | Case C05; Vol I Ch 8, Ch 9; Vol II Ch 15; Vol III Ch 10.                                                                                     |

1

### Collect evidence

- Forecasts, bids, schedules, activations, metering, settlement, constraints, timestamps, and contract rights.

2

### Build the benchmark

- Use only actions that were feasible at the relevant decision time.

3

### Attribute leakage

- Market conditions, forecast error, poor bidding, conservative limits, data problems, and contract design have different remedies.

4

### Escalate

- Escalation is credible when materiality, feasibility, and contractual responsibility are visible.



# C06: where is the boundary of a data-centre clean-power product?

|           |                                                                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What has to be bounded before selling a 24/7-style or hourly clean-power claim?                                                                                         |
| ANSWER    | The seller has to define load shape, deliverable supply, firming, control rights, certificate evidence, residual hard hours, and fallback rules.                        |
| ANCHOR    | Scenario hourly matching rises from 58% toward an 85% threshold; stress-hour firming is about 0.99m EUR/year against a 1.40m EUR/year buyer premium before other costs. |
| DEEPEN IN | Case C06; Vol III Ch 5, Ch 7; Vol II Ch 4, Ch 6.                                                                                                                        |

## Product-boundary test

| Boundary | Required answer                        | Failure mode                     |
|----------|----------------------------------------|----------------------------------|
| Load     | Hourly demand and flexibility          | Annual average hides hard hours  |
| Supply   | Eligible assets by zone and interval   | Correlated or remote shortfall   |
| Firming  | Who covers residual hours              | Uncapped market purchases        |
| Control  | Dispatch, curtailment, and data rights | Promise without operating rights |
| Evidence | Certificates, meters, audit trail      | Claim quality cannot be defended |

# C07: what operating model belongs in year one?

|           |                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Which capabilities belong in build, buy, outsource, audit, or postpone first?                                                                         |
| ANSWER    | Build around recurring decisions that carry material risk and require proprietary judgment; outsource execution only with data rights and benchmarks. |
| DEEPEN IN | Case C07; Vol II Ch 16, Ch 17; Vol III Ch 10, Ch 11, Ch 12.                                                                                           |

## Capability roadmap

| Decision area           | Year-one posture       | Reason                                         |
|-------------------------|------------------------|------------------------------------------------|
| PPA quote stack         | Build internal method  | Pricing logic and risk boundary are strategic  |
| BESS dispatch benchmark | Build/audit            | Partner actions need feasible comparison       |
| Market execution        | Outsource / hybrid     | 24/7 operations need scale                     |
| Data contracts          | Build internal control | Every later model depends on evidence          |
| Risk reporting          | Build lightweight pack | Management needs recurring decision visibility |
| Full trading desk       | Postpone               | Governance and volume threshold come later     |

# How does a reader use a case?

|           |                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What working habit does the casebook train?                                                                                                 |
| ANSWER    | Write a first decision, name the scarce variable, then compare it with the worked artifact and rerun the case after one constraint changes. |
| DEEPEN IN | Cases C01-C07.                                                                                                                              |

1

## First answer

- State the decision in one sentence.
- Name the scarce variable: MW, MWh, connection, credit, liquidity, data rights, cycle budget, or residual-risk tolerance.

2

## Worked artifact

- Compare the artifact with the first answer.
- Track which risk was assigned, priced, bounded, or left unaccepted.

3

## Constraint flip

- Change one assumption and rerun the decision.
- The best cases show which variable actually controlled the answer.

# What does the casebook stress?

**QUESTION** Which failure modes recur across the cases?

**ANSWER** The same themes repeat: product promises can exceed capability, physical resources cannot be spent twice, finance separates revenue quality, and governance needs evidence.

**DEEPEN IN** Cases C01-C07; Vol II Ch 8-Ch 17; Vol III Ch 10-Ch 12.

## Commercial and physical stress

- Product promises can outrun asset capability, evidence, or control rights.
- MW, MWh, grid access, SoC, degradation, and response requirements cannot be spent twice.

## Financial and governance stress

- Bankable revenue, merchant upside, credit exposure, and liquidity risk have different uses.
- Outsourced execution can work only when data rights, feasible benchmarks, and escalation rules exist.